

Sabyasachi Mukherjee, M.Sc.

✉ mukherjee.ssachi@gmail.com
🐙 <https://github.com/sabyasachi-mukherjee>
🌐 <https://sabyasachi-mukherjee.github.io/>
🌐 <https://www.linkedin.com/in/mukherjee-sabyasachi/>



Summary

- Python developer in Berlin. Currently learning the Python backend framework Django!
At the moment, I work on developing software in python as a Machine learning engineer in Berlin. I enjoy hard problems that require my problem solving skills developed during my master's degree in mathematics.

Employment History

1st August 2022 – 31st December 2022

- Machine Learning Engineer (Intern). Smart Steel Technologies GmbH, Berlin, Germany.

Summary

- German job-seeking visa already with permission to work full time : No visa sponsorship necessary.
- Earliest Entry Date: 01.01.2023.
- Salary Expectations: € 48000-50000 per year.

Education

- 2022 M.Sc. Mathematics, Georg-August-Universität Göttingen, Germany.
Thesis title: *The Möbius Function on Arithmetic Progressions.*
- 2017 B.S (Research). Mathematics, Shiv Nadar University, India.
Thesis title: *Dirichlet's Theorem on Arithmetic Progressions.*

Skills

Technical Knowledge	Supervised and Unsupervised Machine learning, clustering, k -nearest neighbours algorithm, Mathematics, Probability and Statistics, Coding, Linux command line, Git : Gitlab and Github.
Languages	English, Hindi, Bengali – Fluent, German – A2.
Coding	Python, SQL, $\text{\LaTeX}$, basic HTML.
Python packages and frameworks	Scikit-learn, Pandas, XGBoost , Seaborn, Matplotlib, SciPy, Numpy, Pytest, Jupyter, Docker, Django.
Relational Databases	PostgreSQL, Clickhouse
Misc.	Academic research, teaching.

Contributions

- Built a proof of concept machine learning model on imbalanced tabular dataset that predicts defects in steel post-casting, beating random forests and XGBoost in terms of recall with similar precision. This classification model was rule-based, informed by insights supplied by deeper analysis of the data set.
- Used libraries like pandas (matplotlib backend) and Seaborn to detect anomolous behaviour, thereby quantifying damage caused by a certain leakage in customer's steel plant.

Responsibilities

- | | |
|------------------|---|
| Data Cleaning | ■ Using pandas to clean real industrial time-series data from steel plants. |
| Visualisation | ■ Using Matplotlib and Seaborn to obtain insights into data |
| Tests | ■ Writing Pytests to test code written by colleagues. |
| Machine Learning | ■ Building proof-of-concept machine learning models that predict defects in certain grades of steel |

Miscellaneous Experience

- | | |
|---------|--|
| Udemy | ■ Course: Machine Learning, Data Science and Deep Learning with Python. Link to Certificate. |
| Hobbies | ■ Floorball and badminton, mathematical research. |